

LA HW

Included exercises: 2, 8, 11, 19, 27, 29, 35, 37, 77-78, 80

Exercises 1-10

In Exercises 1–10, find a generating set for each subspace.

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

Original Exercise 2

$$\left\{ \begin{bmatrix} 2s \\ -3s \end{bmatrix} \in \mathcal{R}^2 : s \text{ is a scalar} \right\}$$

Original Exercise 8

$$\left\{ \begin{bmatrix} r - s + 3t \\ 2r - t \\ -r + 3s + 2t \\ -2r + s + t \end{bmatrix} \in \mathcal{R}^4 : r, s, \text{ and } t \text{ are scalars} \right\}$$

Exercises 11-18

Col A, where A is the matrix used in Exercises 11–18.

$$\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

Original Exercise 11

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & -1 & 1 \end{bmatrix}$$

Exercises 19-26

In Exercises 19–26, determine whether each vector belongs to $\text{Col } A$, where A is the matrix used in Exercises 11–18.

$$\begin{bmatrix} 2 \end{bmatrix}$$

$$\begin{bmatrix} -1 \end{bmatrix}$$

$$\begin{bmatrix} 1 \end{bmatrix}$$

Original Exercise 19

$$\begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}$$

Exercises 27-34

$$5. \left\{ \begin{bmatrix} -s + t \\ 2s - t \end{bmatrix} \in \mathcal{R}^3 : s \text{ and } t \text{ are scalars} \right\}$$

Original Exercise 27

matrix.

$$\begin{bmatrix} -1 & 1 & 2 \\ 1 & -2 & 3 \end{bmatrix}$$

Original Exercise 29

$$\begin{bmatrix} 1 & 1 & -1 & 4 \\ 2 & 1 & -3 & 5 \\ -2 & 0 & 4 & -2 \end{bmatrix}$$

Exercises 35-42

$$\begin{bmatrix} 1 & 1 & -1 \\ 1 & 0 & 0 \end{bmatrix}$$

Original Exercise 35

$$T \left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \right) = [x_1 + 2x_2 - x_3]$$

Original Exercise 37

$$T \left(\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \right) = \begin{bmatrix} x_1 + x_2 \\ x_1 - x_2 \\ x_1 \\ x_2 \end{bmatrix}$$

Original Exercise 77

Prove that the intersection of two subspaces of $\mathcal{R}^n$ is a

Original Exercise 78

Let

$$V = \left\{ \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \in \mathcal{R}^2 : v_1 = 0 \right\}$$

and

$$W = \left\{ \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \in \mathcal{R}^2 : v_2 = 0 \right\}.$$

(a) Prove that both V and W are subspaces of $\mathcal{R}^2$.

Original Exercise 80

and all scalars c , the vector $\mathbf{u} + c\mathbf{v}$ is in $\mathcal{S}$.

Prove that if V is a subspace of $\mathcal{R}^n$ containing vectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k$, then V contains the span of $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_k\}$. For this reason, the span of